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Scaling Spectroradiometric Monitoring to an Algal Factory Farm

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ABSTRACT

This report briefly summarizes the spectroradiometric approach adapted and deployed as part of Task 4.2 within the BETO-funded **D**evelopment of **I**ntegrated **S**creening, **C**ultivar **O**ptimization, and **V**erification **R**esearch (DISCOVER) project, specifically task 4, “Increase crop protection and culture stability.” The goals of this subtask were to monitor ponds cultivating a variety of DISCOVER strains at the Arizona Center for Algae Technology and Innovation (AzCATI) with spectroradiometric technology to:

- Predict biomass concentrations with high temporal frequency.
- Optimize automated analysis methods that enabled feedback for pond automation strategies
- Identify pond crash indicators from primary (pest/predator) and indirect (change in algal host) spectroscopic signals with potential for early treatment/intervention
- Provide recommendations for technology deployment at scale.

This report fulfills the deliverable requirement for the final bullet “provide recommendations for technology deployment at scale”.

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ACRONYMS AND DEFINITIONS

Abbreviation	Definition
ATP ³	Algal Testbed Public Private Partnership
AzCATI	Arizona Center for Algae Technology and Innovation
CCD	Charge coupled device
COTS	Commercial Off the Shelf
DISCOVER	Development of Integrated Screening, Cultivar Optimization, and Verification Research
HDPU	Hyperspec Compact Data Processing Unit
LDRD	Laboratory Directed Research and Development
LED	Light Emitting Diode
OD	Optical Density
qPCR	Quantitative Polymerase Chain Reaction
SOO	Surface of Opportunity
SWaP	Size, Weight, and Power
UAV	Unmanned Aerial Vehicle
VNIR	Visible to Near infrared

1. INTRODUCTION

Outdoor algal ponds are complex living systems where resiliency is vital for productivity, and environmental contamination can significantly reduce output. Estimates suggest that annual production losses from grazers alone may reach 30%, excluding impacts from other pests like fungi, bacteria, viruses, and invasive algae. Even in the ATP³ Unified Field Studies, which utilized mini-raceway ponds elevated on metal frames to minimize environmental exposure, the average time between pond failures due to invaders was only about three weeks. Production losses can also be observed from non-optimal operation parameters and abiotic factors including pH shifts, weather, and non-optical cultivation conditions. Efforts are underway to develop algal monitoring and mitigation strategies for algal culture health, including pest contamination. However, current monitoring techniques, such as optical density and qPCR, are offline methods requiring laboratory access and skilled personnel, making them unsuitable for the rapid, broad coverage needed by the agricultural community. Traditional farming has been moving towards precision agriculture, using remote spectroradiometric sensing data from multispectral and hyperspectral instruments on airborne-type platforms to address the need for near-real-time monitoring of crop health. The spectroradiometric sensing approach leverages the use of biomarkers related to crop health that arise as different wavelengths of light are absorbed and reflected based on the crops' physical and chemical properties.

Starting more than a decade ago, Sandia has built upon the efforts of the oceanography community to develop optical spectroscopy-based methods for monitoring algal health and productivity in the field. The approach, termed spectroradiometric monitoring, leverages the physical and optical properties of algae to enable near real-time biomass measurements and early detection of pests. Publications have demonstrated high temporal frequency (2-5 minutes), quantitative measurements, scalability from small to large environments, identification of deleterious pests 1-3 days before a crash occurs, and potential for autonomous operation. However, while the method shows promise for improving algal crop protection and control and has been fielded autonomously during DISCOVER on multiple raceway ponds for at least 3 years, it may need to circumnavigate challenges related to scale up for a larger algal factory farm.

This report summarizes the technology and challenges associated with the current point spectrometer-based format (Section 2), presents technological solutions for addressing the scale-up (Section 3) and provides a vision for future scale up needs (Section 4).

2. REVIEW OF SANDIA'S EFFORTS TO DEVELOP SPECTRORADIOMETRIC METHODOLOGY

Past efforts at Sandia have employed fixed position, single point spectrometers to measure algal biomass and pest presence. This approach is well-suited for well-mixed raceways ponds. Sandia's spectroradiometric monitoring effort began with internal funding through the Laboratory Directed research and Development (LDRD) program and has evolved significantly since its original laboratory demonstration over 15 years ago. A concise summary of that evolution is provided below, focused on our data collection and processing approaches which have employed fixed position, single point spectrometers to measure algal biomass and pest presence. The final section summarizes the current assumptions and ongoing challenges of implementing the approach in the field.

2.1. Data collection

The original embodiment of the spectroradiometric monitoring approach, developed and demonstrated under Sandia's *Benchmark to Raceway* LDRD project (2010-2012), was focused on assessing biomass a laboratory culture of *Nannochloropsis salina* grown under fluorescent lighting (1). It is worth noting that the original system architecture, which incorporated an upward-staring fiber with a cosine-corrector attachment to capture the downwelling irradiance, a downward-staring bare fiber to capture the upwelling radiance from the culture, and then calibrating the wavelength-resolved reflectance to a known diffusion reflectance target, has stood the test of time and is still utilized in the DISCOVER deployments at AzCATI.

Initial field deployment occurred at Sapphire Energy (Las Cruces, NM), involving a 4-week monitoring of two ponds, one culturing green algae (of the genus *Desmodesmus*) and the other cyanobacteria (of the genus *Arthrospira*). Both the laptop computer and spectrometers were enclosed in an outdoor refrigerator regulated by a temperature control unit. Data were acquired continuously without user intervention for the entire 4-week span and biomass was successfully predicted using the spectroradiometric approach in 5 minute intervals (2). The only issue involved our deployment of the same laboratory-grade optical fibers that had been used for monitoring our laboratory culture. When exposed to the higher irradiance from direct sunlight the longer lengths of these fibers resulted in an unwanted background due to light leaking through the outer cladding. Field-grade fibers jacketed with stainless steel monocoils to eliminate light leakage were thus used in all subsequent outdoor deployments.

Since the initial 2012 outdoor demonstration at Sapphire Energy, our field work has focused on deploying the monitoring technology at AzCATI, first under ATP³ (Algae Testbed Public Private Partnership, 2013-2016), and subsequently under DISCOVER (2017-2025). The approach remained relatively unchanged from the original Sapphire Energy deployment with only minor updates to equipment as needed. The DISCOVER project enables an extension of the effort from biomass prediction to detecting pest presence in outdoor raceways (3). Recently, the frequent power outages at AzCATI, which require a manual restart of the computer software, led to development of a more autonomous implementation using a Raspberry Pi unit to control data acquisition for seven spectrometers simultaneously and enable automatic restarts after recovering from power outages. Even more recent has been the addition of real-time analysis pipeline which eliminates facilitates feedback control and could be employed for real-time decision making about pond operation. Both advancements are being reported in papers under consideration for the upcoming DISCOVER special issue in *Algal Research* (4, 5).

2.2. Data processing

2.2.1. The single-backscattering albedo

Our physics-based model parameterizes the reflectance in terms of the algal absorption a and backscatter b_b , a parameterization that has been used within the oceanography community for over the past five decades (6). Specifically, our model expresses the reflectance as a function of the unitless single-backscattering albedo u , defined as the ratio of the backscatter b_b to the sum of the absorption a and b_b ,

$$u = \frac{b_b}{a + b_b}.$$

2.2.2. Impact of multiple scattering

The single scattering approximation (SSA) is typically employed for analysis of radiative transfer in tenuous media. In this approximation, one neglects each particle's response to fields scattered by other particles (7), meaning that each particle's contribution to the reflectance is defined solely by its interaction with the incident field, which results in the reflectance scaling *linearly* with the single-backscattering albedo. But due to multiple scattering, the relationship observed in dense algal cultures is instead *nonlinear*, the exact nature of this relationship being defined by radiative transfer within the culture. Here we leverage the work of Lee et al. (1999), who quantified the relationship to u by implementing numerical solutions via Hydrolight (<https://www.sequoiasci.com/product/hydrolight/>), a radiative transfer solver tailored for application to bodies of water. Assuming a scattering phase function typical of coastal waters, Lee et al. (1999) determined the below-water-surface reflectance for an optically deep culture, r_{bs}^{od} , to be well-approximated as a simple quadratic function of u ,

$$r_{bs}^{od} \approx (0.084 + 0.17u)u,$$

with the coefficients of 0.084 and 0.17 determined via Hydrolight simulations.

2.2.3. Extending to optically thin cultures

When accounting for cultures that are not optically deep, the below-water-surface reflectance corresponding to a below-surface solar zenith angle of θ_w was determined by Lee et al. (1999) to be

$$r_{bs} \approx r_{bs}^{od} \left(1 - \exp \left\{ - \left[\frac{1}{\cos(\theta_w)} \right] + 1.03(1 + 2.4u)^{0.5} \right\} \kappa H \right) + \frac{1}{\pi} \rho \exp \left\{ - \left[\frac{1}{\cos(\theta_w)} \right] + 1.04(1 + 5.4u)^{0.5} \right\} \kappa H \right)$$

in which the coefficients of 1.03, 2.4, 1.04, and 5.4 were again determined via Hydrolight simulations. In the above equation, the optically deep reflectance r_{bs}^{od} has been multiplied by an expression to account for the truncation of the light path owing to the bottom surface, while a second term has also been added, which accounts for contribution by light reflected by the bottom-surface of reflectance ρ to the total reflected light. The term κH is the product of the quasi-diffuse attenuation coefficient ($\kappa = a + b_b$) and the water depth H . As κH increases, the culture becomes more optically deep, with r_{bs} approaching r_{bs}^{od} .

2.2.4. The above-surface reflectance

We note that the above equation solves for the below-water-surface reflectance, as would be measured by a probe positioned just below the water surface. For our spectroradiometric monitoring approach, the probe instead detects light above the water surface. The refractive index difference between water and air results in an internal reflection at the water-air surface, thus reducing the light that escapes the interface, such that the above-water-surface reflectance r_{as} becomes

$$r_{as} \approx \frac{0.5r_{bs}}{1 - 1.5r_{bs}},$$

with the coefficients of 0.5 and 1.5 once again derived via Hydrolight simulations. The above equations provide an end-to-end relationship between u and the reflectance resulting from light scattered and absorbed by the algae, to which our model adds the impact of algae fluorescence, as well as the reflection of sunlight and skylight from the water surface, as described by Reichardt et al. (2014).

2.2.5. Wavelength-dependent sensitivity of reflectance to algal biomass

Considering wavelengths for which algae (vs. water) is the dominant absorber, as is the case for visible light (400-700 nm), both a and b_b should increase approximately linearly with biomass, meaning that u will be insensitive to changes in biomass. As a result, as soon as the culture becomes optically deep, the reflectance over the visible region will remain unchanged as the biomass continues to increase. Thus, a collection system limited to visible wavelengths would be of very little use for assessing algae biomass. On the other hand, at wavelengths for which a is dominated by the water absorption water (>750 nm), only b_b scales linearly with biomass, resulting in u being sensitive to changing biomass, so the reflectance remains a function of biomass, even for optically deep cultures.

2.3. Assumptions

The assumptions associated with spectroradiometric monitoring approach we have developed can be divided into two categories: assumptions that are inherent to the model and assumptions that are based on the implementation of the technology. With respect to the former category, the primary assumption is that the irradiance incident upon the pond surface is the same as that measured by a cosine-corrected sensor pointed in the upward direction. That assumption breaks down in situations such as when shadows occlude incident light from the pond surface or if sunlight is reflected from a side wall and onto the surface of the raceway pond making a secondary illumination source and the model cannot appropriately account for these phenomena. Additionally, because our data acquisition inherently averages over time, whether this averaging occurs over just the integration time of a single acquisition, or over multiple such acquisitions averaged together, temporal variations can occur during any given averaging window within the upwelling sensor field-of-view – e.g., a clump of flocculated culture floating past the sensor – or within the downwelling sensor field of view – e.g., localized cloud cover or fast moving clouds.

When optimizing the model to measurements, we typically assume that the bottom surface reflectance remains constant. This assumption would be violated if the culture began settling toward the bottom. Related to the bottom surface reflectance, the models of Lee et al. (8, 9) assume that it is Lambertian, an assumption inconsistent with the smooth surfaces of small-scale raceways, and especially poor when applied to the glass bottom of a beaker or flask. There are a host of additional assumptions Lee

et al. (8-10) invoked in the development of their reflectance models, described by their respective papers, to which we refer the reader.

Beyond the assumptions that effect the model, we also have implemented a point-based monitoring approach that inherently assumes the pond is well-mixed and any given point we monitor is representative of the pond. While this assumption might be valid for small, research type raceways, modeling has indicated the presence of unmixed “dead zones” in open raceway ponds especially as the pond sizes increase and paddlewheel efficiency becomes limited (11-14). Although open raceway ponds serve as the principal system for large-scale microalgae cultivation, accounting for 90% of global microalgae production (15), alternate cultivation systems do exist where concentration variations are expected due to minimal or lack of mixing such as natural bodies of water, shallow unstirred ponds, incline ponds, and attached algae systems (e.g. algal turf scrubbers (16)).

Together these assumptions serve as strong motivation for implementing spatially-resolved sensing approach, to evaluate the assumption of a well-mixed pond, and potentially enable the detection of variations of biomass within a pond and/or multiple ponds across a large area. Spatially-resolved sensing could be accomplished through either a spectral imaging-based approach enabling the assessment of many spatial locations simultaneously using a 2D imaging sensor or a scanning approach where a point spectrometer is scanned over an area to build up an image. Alternatively, a combination of both using an imaging detector and scanning it could achieve larger coverage. The technology behind these approaches will be presented in the next section.

3. EXTENDING SPECTRORADIOMETRIC METHODOLOGY TO THE ALGAL FARM FACTORY

3.1. Introduction to spatially-resolved spectroradiometric approaches

Hyperspectral reflectance measurements can be collected in a spatially-resolved manner to address the limitations described above for non-uniformly mixed algal cultures. The goal would be to record a three-dimensional (3D) hyperspectral data cube consisting of 2 spatial dimensions and one spectral dimension at various points in time. There are several data collection strategies to achieve this, each with distinct characteristics and trade-offs in terms of spectral, spatial, and temporal resolutions. The three types relevant for this application are depicted in Figure 1.

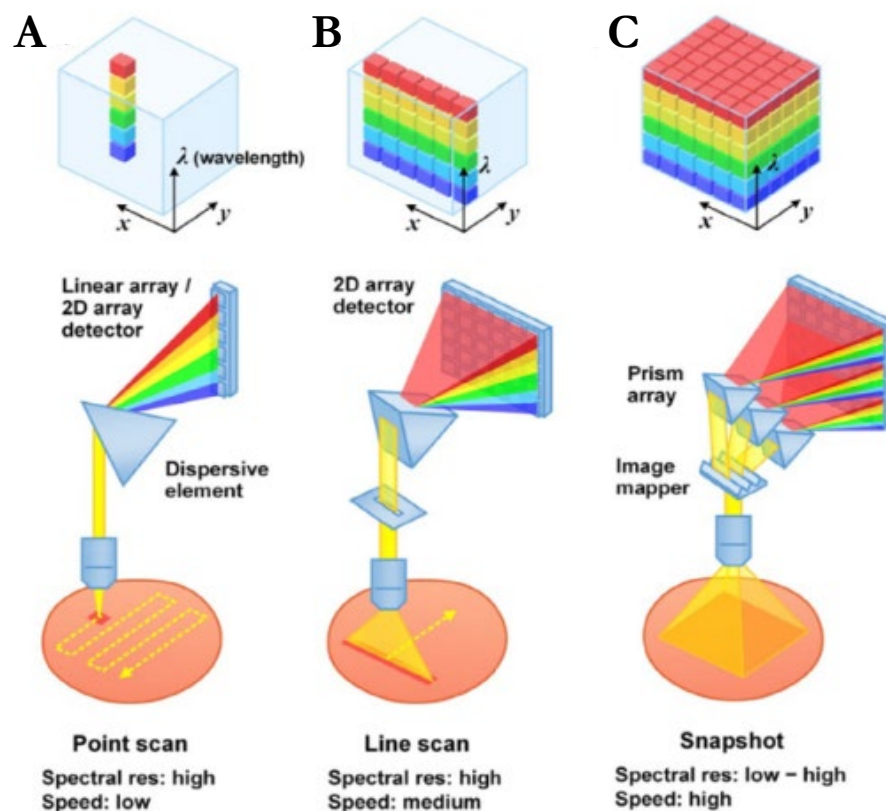


Figure 1: Architectures for collecting spectrally resolved reflectance measurements. Figure has been modified from <https://doi.org/10.1002/wnan.1661>.

3.1.1. Single point spectrometers

Single point spectrometers such as the ones deployed at AzCATI in the DISCOVER project capture spectral data at a single spatial location typically through a fiber optic and pass this through a dispersive element, such as a prism or diffraction grating and onto a detector such as a charge coupled device (CCD) detector or a linear array. To collect spatially-resolved information over an entire scene, these single point spectrometers would need to be moved from point-to-point to build up the image of the scene (Figure 1a). While the single point spectrometers approach offers high spectral resolution and

sensitivity, the major limitation is the time required to scan an entire scene, making the approach less suitable for heterogeneous scenes such as non-uniformly mixed algal raceways or slow flow, non-flow environments. Even if a single point spectrometer was mounted on an unmanned aerial vehicle (UAV) and flown over the raceways, the time to collect an entire image would likely be too long relative to the time it takes the algae to traverse the raceway. A UAV-mounted single point spectrometer could potentially be utilized in a non-flow or slow flow cultivation configuration.

3.1.2. *Line scan hyperspectral imagers*

Line scan hyperspectral imagers (also known as push broom hyperspectral imagers) capture spectral information along a single line of the scene at a time, using a 2D detector. Then the line is scanned along the scene to build up the image (Figure 1b). This approach balances spatial and spectral resolution, making it suitable for applications like aerial environmental imaging and industrial inspection, including applications in algal monitoring for harmful algal blooms (17). Recently the exquisite spectral resolution of the line scan hyperspectral imaging approach was demonstrated in the laboratory to assess biomass in different microalgal strains (18) and discriminate five cyanobacterial genera that can cause harmful algal blooms (19). While line scan hyperspectral imagers offer significantly faster data acquisition than single point spectrometers because an entire line of spatial points are collected simultaneously, a line scan hyperspectral imager can still face limitations in capturing rapidly changing scenes due to the scanning architecture. The most optimal configuration is achieved when the scanning direction is correlated with the flow within a scene. In aerial imaging for terrestrial and environmental applications line scan hyperspectral imagers are typically mounted on an aircraft or UAV and the flight pattern creates the scanning direction although gimbal mounts can also be used to sweep the scene and build up the image if the imager is stationary.

3.1.3. *Snapshot hyperspectral imagers*

Snapshot hyperspectral imagers capture the entire 3D hyperspectral data cube in a single exposure, allowing for instantaneous measurements across a scene. Utilizing advanced optical designs covered in more detail in Section 2.2 of the recent review by Yako (20), these systems can excel in speed, making them ideal for dynamic applications and they typically have low SWaP making them easily deployable in the field. However, they generally possess lower spectral resolution (1.5 – 2 orders of magnitude lower), reduced sensor sizes compared to single point spectrometers and line-scan hyperspectral imagers, and lower signal-to-noise ratio due to simultaneous wavelength capture. Nevertheless, snapshot hyperspectral imagers are becoming increasingly popular for agricultural applications especially when combined with machine learning and dee learning approaches (21-24).

Table 1 summarizes the unique strengths and weaknesses of each hyperspectral technology with respect to large area scanning of algae farms.

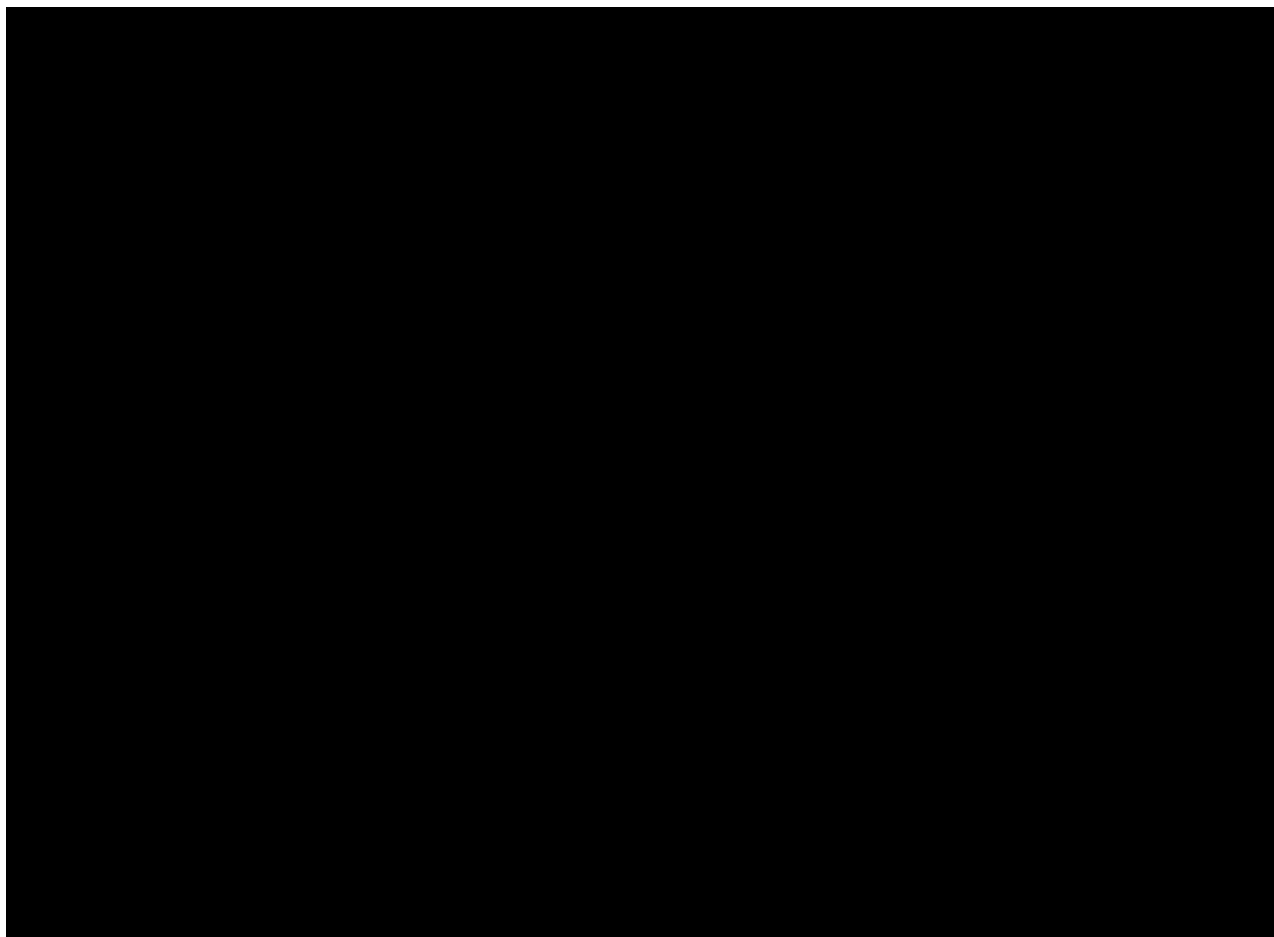
Table 1: Summary of advantages and disadvantages of three different hyperspectral reflectance technologies

Technology	Advantages	Disadvantages
Point Spectrometers	• Inexpensive (~\$1-5K) • Easy to use and transport • High spectral resolution	• Can only “see” a small area, no spatial resolution
Snapshot Hyperspectral Camera	• Spatially-resolved • Fast	• Low spectral resolution • Limited Field of View

Technology	Advantages	Disadvantages
	• Easy to use and transport • Moderate cost (\$12-25K)	
Line-scan Hyperspectral Camera	• Spatially-resolved • Can scan a large area • High spectral resolution	• Expensive (\$75 – 100K) • Large/heavy

3.2. Lab demonstration

An experiment was designed and conducted to measure algae at different concentrations in the laboratory setting with all three of the technologies described in Table 1. Algae (*P. celer* TG2 & *N. gaditana*) was grown at room temperature ($\sim 25^{\circ}\text{C}$) on an orbital shaker rotating at ~ 120 rpm) under white LED lights ~ 140 μM with a 12/12 light/dark cycle. *P. celer* TG2 was grown in DISCOVER 35 marine media – a modified version of F/2, while *N. gaditana* was grown in BG-11 media. Algal cultures were maintained in stationary phase and on the day of the experiment, dilutions were prepared by the addition of media to span the range from 0.1 – 0.8 OD. Approximately 300 ml of each concentration was placed into clean crystallization dishes (Cole Parmer, 100 mm x 50 mm) and the dishes arranged in the field of view of the hyperspectral imagers as shown in Figure 2 and Figure 3.



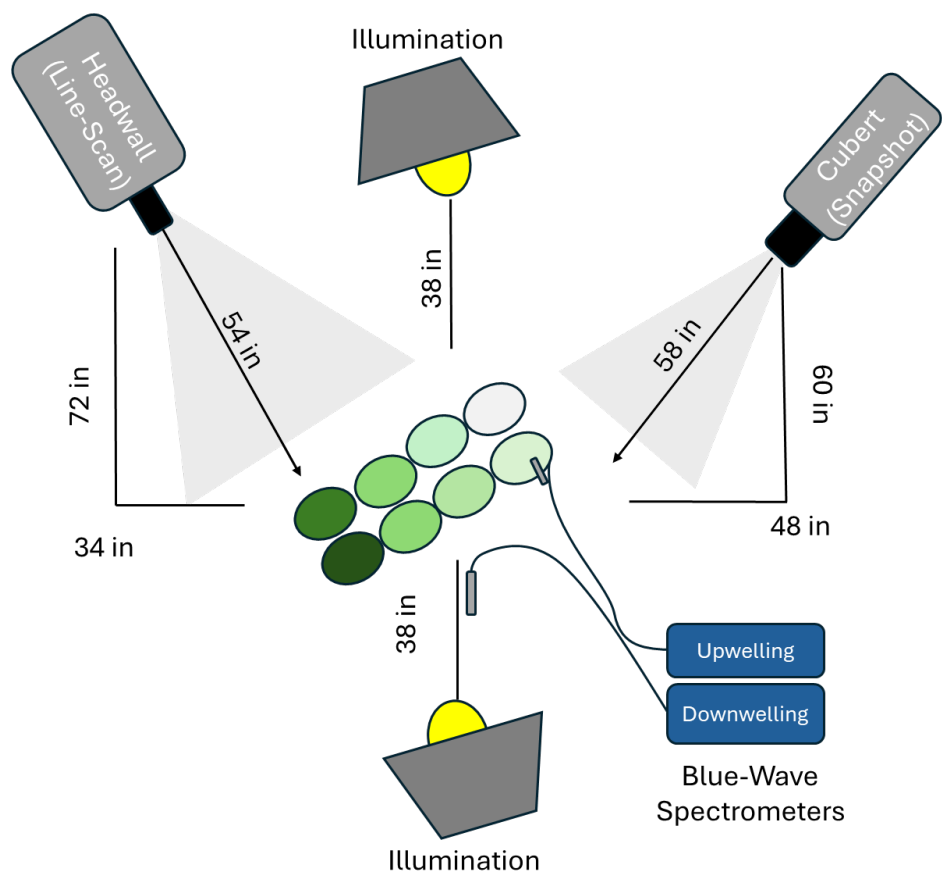


Figure 2: Schematic of laboratory experiment set up

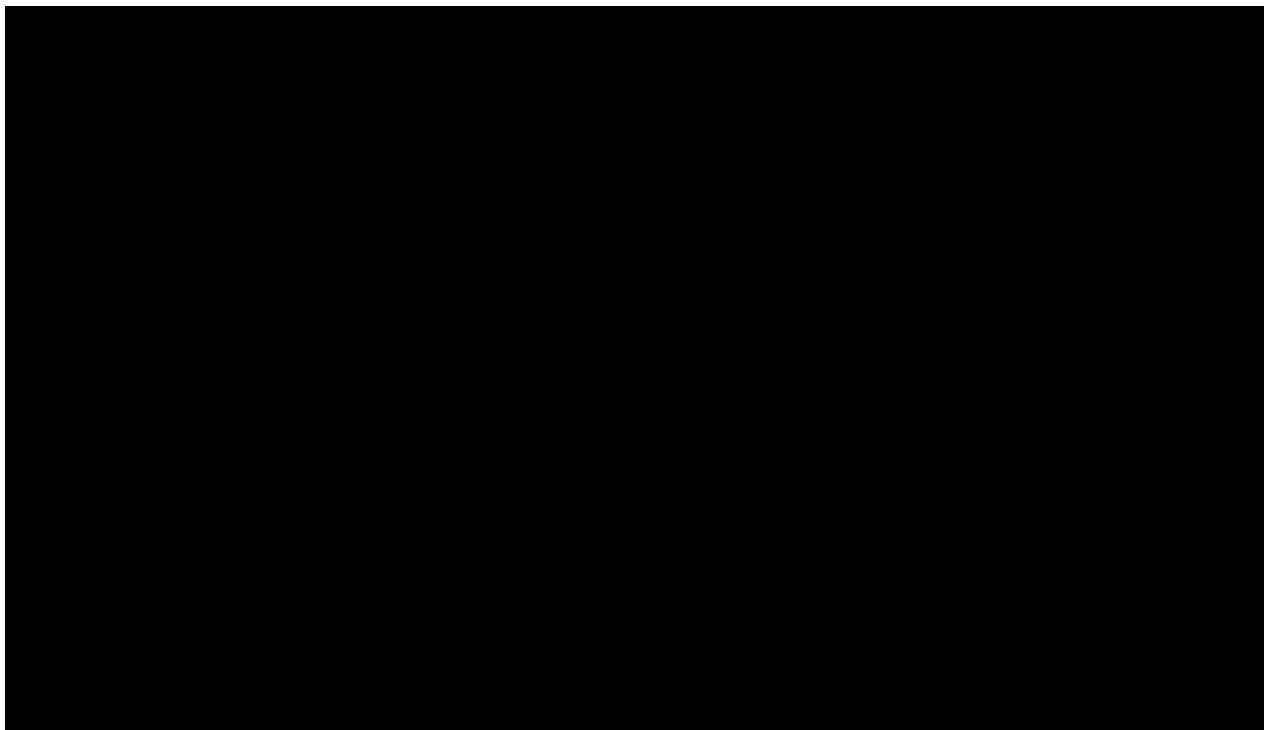




Figure 3: Photograph of lab experiment showing dishes with various *P. celeris* TG2 concentrations. OD readings are indicated on each dish.

Spectrally resolved reflectance spectra were processed by subtracting dark counts and converted to reflectance values by ratioing to the diffuse reflectance target (also with the dark counts subtracted). Hyperspectral image data (both snapshot and line-scan format) was converted to reflectance values on a per-pixel basis using the same method, performing a ratio to data collected from a diffuse reflectance target under the same conditions. Figure 4 shows representative spectra (top panel) and spectral images (bottom panel) from a *P.celeris* dilution experiment. The spectral shapes are similar from

can achieve read-out rates of 200-300 lines/sec which is $\sim 60\times$ faster than the 5 lines/sec used in this experiment.

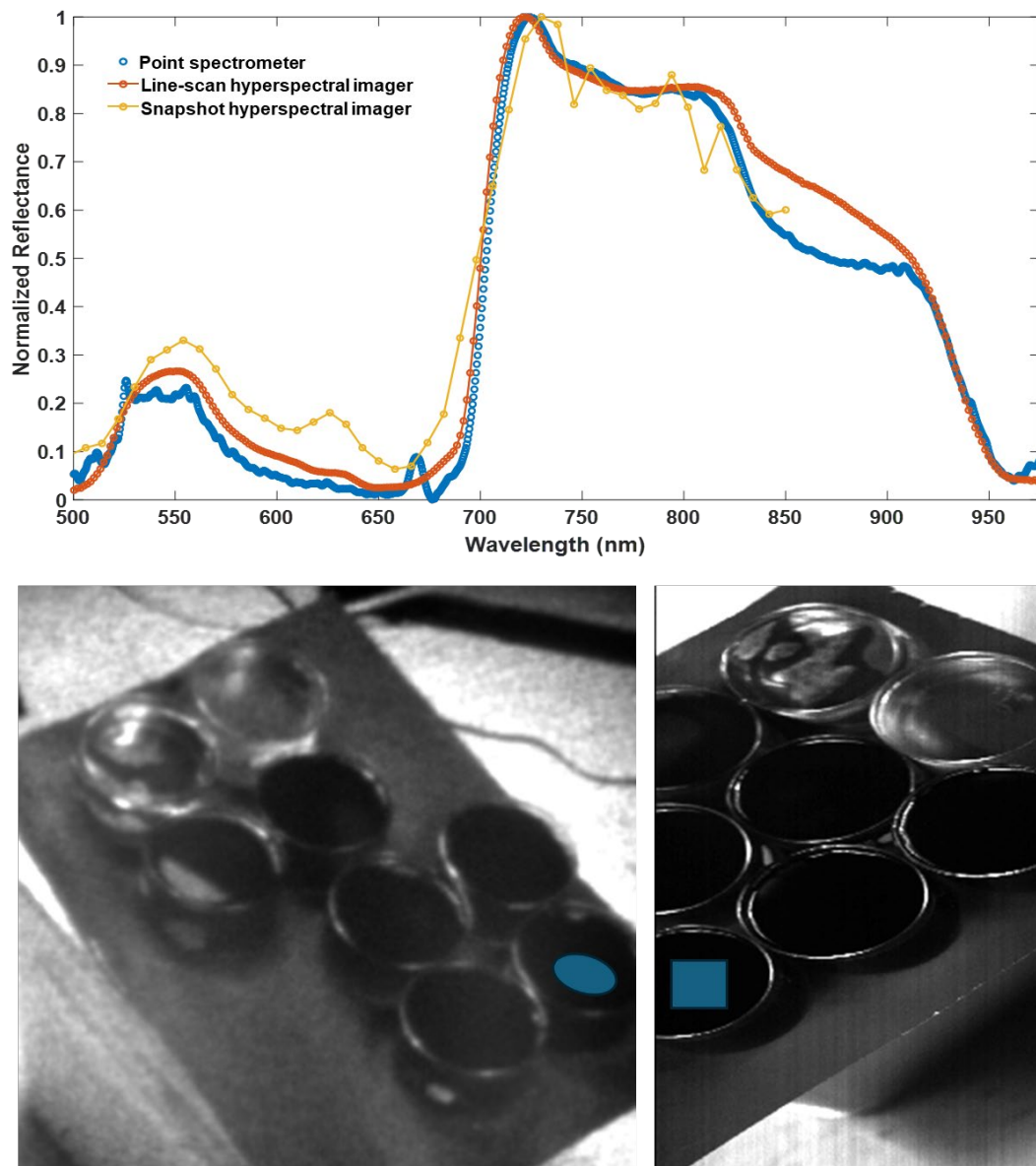


Figure 4: Laboratory data from *P.celeri* dilution experiment. Top Panel: Comparison of reflectance spectra from “Dish 8” (highest algal concentration, an OD_{680nm} of 0.58) for all detectors highlighting trade-off in spectral resolution and coverage with the different detection modalities. Point spectrometer trace is the average of 25 temporal scans at a single spatial location, while the line-scan and snapshot hyperspectral traces are averages of 62500 and 703 spatial pixels, respectively. Bottom left panel: Single wavelength image (650 nm) from snapshot hyperspectral camera of dishes with various algal concentrations. Oval area highlights the region from which the average spectra in top panel (yellow trace) was obtained. Bottom right panel: Single wavelength image (650 nm) from line scan hyperspectral camera of dishes with various algal concentrations. Square area highlights the region from which the average spectra in top panel (red trace) was obtained. Note: the dishes were rearranged slightly between the snapshot and hyperspectral images to accommodate the different field of view.

3.3. Field demonstration

On October 2, 2025, passive hyperspectral image data was collected on Pond 10 at AzCATI. The pond contained *Scenedesmus obliquus* UTEX393 with an $\sim$ OD of 0.715. The paddlewheel was operating normally, and no operational parameters were modified for the experiment. A snapshot hyperspectral imager (Cubert Ultris H5) equipped with the relay lens option and a c-mount lens (Headwall Photonics, 25mm F/L Telecentric VNIR Lens, f/2.5) was aimed at Pond 10 at a standoff distance of $\sim$ 30 inches. This standoff distance was chosen based on convenience (no need for ladders and extension cords), rather than any optimized coverage of the pond. Hyperspectral acquisition was accomplished using the CUVIS UltrisTouch software (v2.9.1) running under Windows 11 on a HP Elitebook 1050 G1 with an Intel(R) Core™ i7-8850H CPU @ 2.60GHz processor, 16 GB of RAM, and an NVIDIA GeForce GTX 1050 graphics card. The arrangement is shown in Figure 5. Hyperspectral data were acquired of the scene with an integration time of 50 ms and 10 images were averaged. Calibration data were collected using a reflectance target with $\sim$ 20% reflectance with the exact acquisition parameters as the pond spectral data. The resulting hyperspectral images were collected 275 x 290 pixels x 51 wavelengths from 450 – 850 nm. This results in contains 79,750 individual spectra. Each pixel's size in terms of area sampled on the pond is dependent on the lens and standoff distance.



Figure 5: Picture of snapshot hyperspectral camera recording data from Pond 10 at AzCATI.

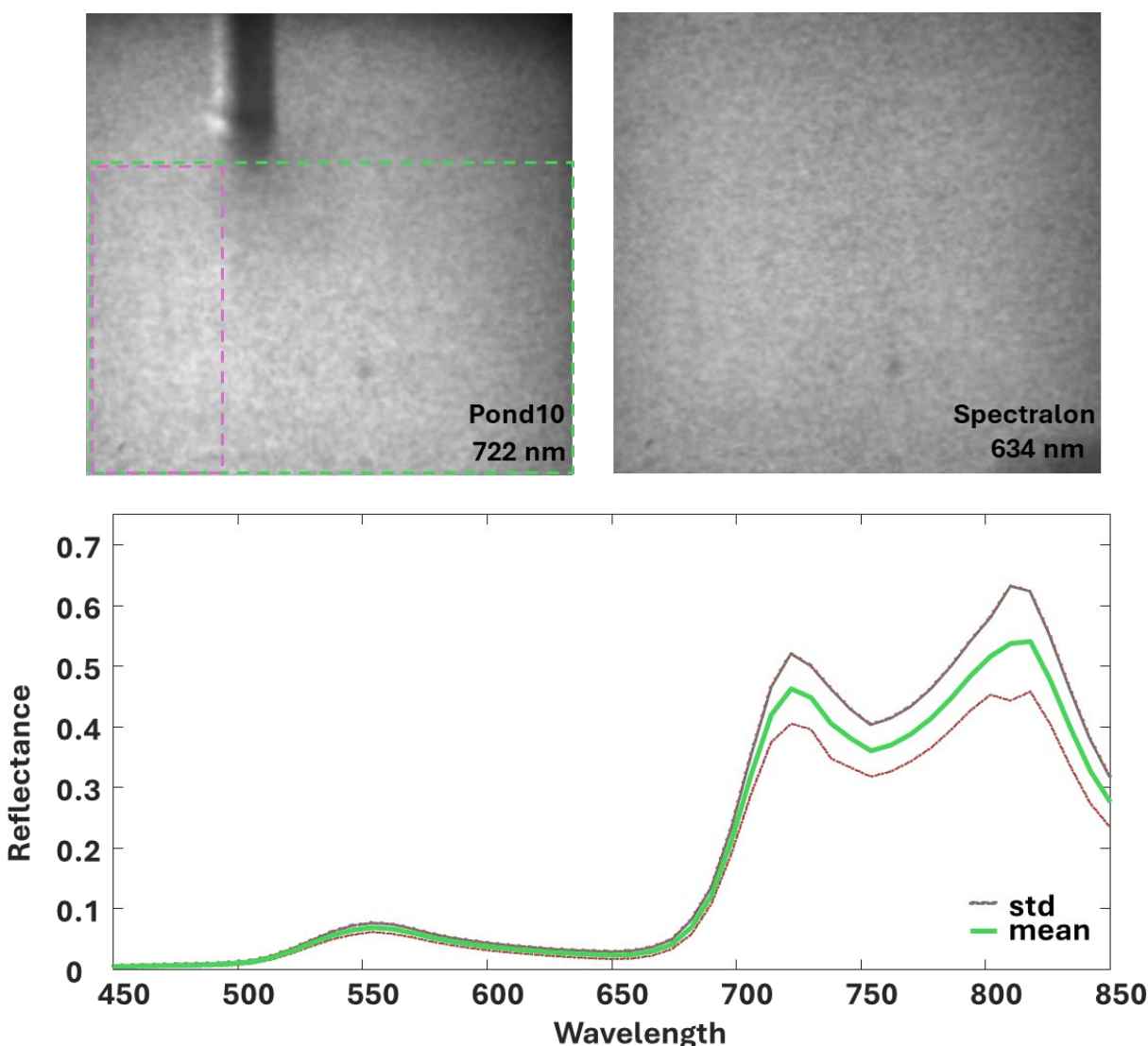


Figure 6: Top Panel: Representative single wavelength images for Pond 10 (left) and Spectralon® reflectance target (~ 20% reflectance, right). Images are 275 pixels x 290 pixels. Bottom: Mean and standard deviation of the 51,040 reflectance spectra within the region of interest indicated by the green dashed rectangle.

A representative single wavelength image from the hyperspectral data cube for both the algae and the Spectralon® reflectance target is shown in Figure 6, top panel, while the bottom panel highlights the variation within a region of interest containing >50,000 spectra. While the variation could be the result of slight variations in algal concentration in the area imaged (the pole providing instrument access could theoretically disrupt the algal flow and prohibit complete mixing), it's noted that the amplitude of the standard deviation appears to follow Poisson statistics and is likely dominated by the noise of the measurement.

An inversion of the relative intensities of the peaks at ~720 nm and ~820 nm was observed in this data that has not previously been seen in laboratory or outdoor single point spectrometer data (upper panel of Figure 4 compared to lower panel of Figure 6. While we acknowledge that Figure 4 and Figure

6 are measuring the reflectance spectra of different algae species, the two species have demonstrated similar reflectance spectra in the past. The inversion of relative intensities is more likely a result of detector saturation in the reference target image. Under these collection conditions (5 ms acquisition) the downwelling sunlight was enough to saturate the detector using the 20% diffuse reflectance target in some pixels (~25%). Additionally, saturation was observed occasionally when measuring the algae reflectance as well (~18%). Detector saturation would potentially be reduced at longer standoff distance, but it is possible that if the system were to be deployed additional filters may be necessary to eliminate the potential for saturation. Saturation renders the data non-linear, and it is impossible to predict accurate concentrations under saturated conditions.

A comparison of the estimated biomass was made between the single point spectrometer measurements of Pond 10 and the region of interest in snapshot hyperspectral image in Figure 6. For the analysis of the single point spectrometer data, ASHARP analysis software was run with the typical parameters for predicting biomass used in operation during DISCOVER project. We used the scatter-free absorbance spectrum of *Tetraselmis striata* as a proxy for the scatter-free absorbance spectra of *Scenedesmus obliquus* UTEX393. At a timepoint corresponding with the time of the snapshot hyperspectral acquisitions, the C_a value from the spectrometer data (which we have shown correlates to biomass in prior work) was determined to be 98.2. A version of the ASHARP analysis software was adapted to work with the spectra resulting from the hyperspectral image data cube. This yielded a C_a value of 166.8 on the mean spectrum from the region of interest indicated by the green dashed rectangle in Figure 6. Figure 7 shows an example of the spatially-resolved C_a values obtained from the region of interest indicated by the magenta dashed rectangle in the image in Figure 6. The discrepancy in C_a value between the two methods will be explored in the future. Initially we believe that it could result from the different light collection geometries of the two methods coupled with the non-linear response of the hyperspectral camera at intensities near-saturation. Both issues could be remedied in future hyperspectral imaging efforts through improved calibration and/or adjustment to the

prediction parameters to better represent the geometries. Unfortunately, this would require additional effort that is outside the scope of the current project funding and timeline.

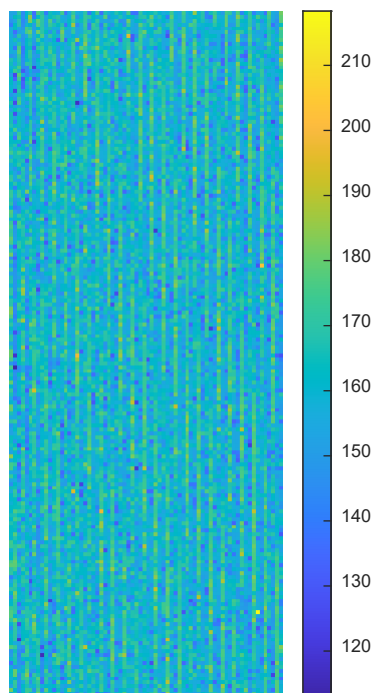


Figure 7: Spatially-resolved C_a values obtained from hyperspectral image shown in Figure 6

4. FUTURE VISION

Spectroradiometric monitoring has found significant application in precision agriculture, with sensors having been deployed on agricultural drones (25) as well as on farm equipment itself (26). Regarding the application to algal factory farming, we envision that these farms will consist of arrays of closely packed, large raceway ponds. As described in Section 2b, while developing our spectroradiometric monitoring approach, we have provided each pond a dedicated sensor to monitor its upwelling radiance. But because deploying a dedicated sensor for each pond of an algae factory farm could be cost-prohibitive, configuring a single sensor to monitor an array of ponds is considered fundamental to the economic viability of spectroradiometric monitoring. Additionally, in larger raceway cultivation strategies the assumption for a well-mixed system could break down, making a single sensor per pond highly inaccurate. Below we consider two sensing platform options for monitoring an array of ponds with a single sensor capable of providing spatially-resolved information on algal biomass status— (1) ground and (2) aerial – as well as share some of the inherent benefits imaging sensors have over point sensors as applied to algal monitoring.

4.1. Ground monitoring units

Rui et al. (2024) review various ground-based agriculture monitoring platforms (see Figure 2), including cable-suspended systems, fixed and movable phenopoles, and manual, semi-automatic, and robotic phenomobiles, such as that developed under the ARPA-e TERRA program (<https://arpa-e.energy.gov/programs-and-initiatives/view-all-programs/terra>)(27). The relative merits of each of these options would depend upon the specific configuration of the algal factory farm. Considering fixed monitoring stations, for example, an elevated imaging sensor could be configured to monitor multiple raceway ponds. A specific embodiment of such an approach would be crane-mounted imaging sensors, which have demonstrated broad spatial coverage for use in, e.g., construction site monitoring (28, 29). Mounting an imager on a multi-axis gimbal would enable scanning the field of view over a broad area, enabling coverage of an array of ponds with a single imaging sensor.

4.2. Aerial monitoring units

For mobile monitoring units, we would consider the use of either a drone-mounted single-point spectrometer or imaging sensor, which could traverse an array of ponds at routine time intervals. An algal factory farm, expected to lack tall structures and provide a clear line-of-sight between the drone and controller, would represent an ideal environment for drone deployment, while specific deployment strategies could leverage drone-based approaches demonstrated for bathymetric and photogrammetric measurements in coastal zones (30). The selection between a point spectrometer or an imaging sensor on the drone would likely be made based on the required spatial coverage, spatial resolution, and temporal resolution. An imaging sensor on a drone would provide an array of images across the flight path with the highest spatial and temporal resolution, but likely has the highest capital cost to implement and represents a significant payload for the drone. A point spectrometer on a drone would provide a more limited spatial coverage in the same time frame as an imaging sensor would, however, can be a factor of 5 -10 lower in cost with lower SWaP making drone flight more achievable.



Figure 8. Examples of ground-based monitoring platforms (Rui et al., 2024)

4.3. Exemplar assessment of ground and aerial monitoring

A comprehensive comparison of ground versus mobile sensing platforms is outside the scope of this report. Nevertheless, here we present an exemplar comparison, which should serve to guide subsequent fuller analyses. This illustrative comparison incorporates the Cubert snapshot sensor, given that this sensor has been deployed in the field, resulting in integration and averaging numbers on which to base the comparison. The specification sheet for the Cubert indicates the maximum resolution of the Cubert sensor is 290 x 275 pixels (we will base this comparison on 275 pixels) while the FOV is 15 deg. We will begin by assuming that a spatial resolution of 0.25 m is required to resolve the gradients due to unmixed regions of large-scale ponds, and we will also assume that the stationary

sensor is positioned on a rotating mount with variable tilt, situated on an elevated platform $h = 10$ m from the ground, imaging a location at a horizontal distance r away from the platform. Normal to the direction of the sensor, the spatial resolution is given by

$$Resolution_{normal} = 2 \left(r^2 + h^2 \right)^{1/2} \tan \left(\frac{FOV}{2 \times 275} \right).$$

Given the pointing angle of the sensor, the spatial coverage in the pointing direction will be elongated relative to that of the direction normal to it, resulting in a spatial resolution in the pointing direction given by

$$Resolution_{pointing} = \frac{Resolution_{normal}}{\sin \left(\tan^{-1} \frac{h}{r} \right)}.$$

For $FOV = 15$ deg and $h = 10$ m, the required 0.25-m resolution in the pointing direction constrains the maximum value of r to 50 m, so the monitored area will extend to a 50-m radius circle around the mounting position. At the perimeter of this circle, the resolution in the direction normal to the sensor pointing is 5 cm, so 275 pixels on the camera sensor will cover a 14 m area on the ground. Given that the perimeter of the 50-m radius circle is 314 m, the entire edge of circle can be covered with 24 angular positions, but multiple such rotations will be required to “fill in” the area of circle. We estimate that 5 such rotations would be sufficient, while also allowing for some spatial overlap. Computing the time required to cover the area, we assume that the rotation mechanism will require 1 s to transition from one angular position to the next. We will also assume that the sensor will require the same integration time (50 ms) and averaging (10 acquisitions) used in the field demonstration, so the time required to cover the entire circular area will be

$$5 \text{ rotations} \times (24 \text{ angular positions} / \text{rotation}) \times (0.050 \text{ s/acquisition} \times 10 \text{ acquisitions} + 1 \text{ sec position-to-position}) = 3 \text{ min.}$$

For comparison, we also calculate the corresponding time for an aerial point sensor traversing the same circular area such as if the single point spectrometer was mounted on a UAV/drone. We treat this sensor very simplistically, assuming the mobile platform can traverse the FOV required for 0.25 m resolution in the 50-ms integration time, so coverage of a 50-m radius area would require

$$\{ [\pi \times (50 \text{ m})^2] / [\pi \times (0.125 \text{ m})^2] \} \times 0.050 \text{ s} = 133 \text{ min.}$$

As was mentioned when introducing this exemplar comparison, there are many assumptions and simplifications incorporated into the above numbers – so much so that they may well be off by an order of magnitude. For example, an aerial point sensor with efficient collection optics may require only a 5-ms integration time, which would reduce the aerial coverage time by a factor of 10. Similarly, regarding the stationary sensor, as r is decreased (e.g., when filling in the circular area), the number of discrete angular positions required will also decrease, also reducing the acquisition time. But also ignored is the impact of the imaging sensor’s limited depth-of-field, and how any blurring that results from this would decrease the spatial resolution. Nevertheless, the above comparison provides a starting framework for assessing the time required to cover an area of ponds, while also providing insight regarding the sensitivity of the resulting value to the input parameters, to benefit future studies.

4.4. Estimation of effort to transition Sandia's DISCOVR efforts to a spatially-resolved platform

Considering the limitations of our point-sensing system discussed in Section 2b, spatially-resolved sensing enables recognizing the presence of shadows, secondary illumination, and importantly any nonuniform culture concentrations (e.g. flocculating culture, imperfect mixing, attached algal systems, etc.) allowing for filtering out the specific areas impacted by these effects. We note, however, that prior attempts to demonstrate transferability between the laboratory and the field (e.g., (31)) do not account for the significant differences between monitoring optically thin petri dishes versus optically deep photobioreactors and raceway ponds. In contrast, our rigorous physics-based approach, developed from decades of work within the oceanography community, and proven out over years of field deployment with point-based sensors, should enable such transferability, toward quantitative assessment of the spatial variability of large-scale raceway ponds.

While the data contained in this report demonstrated proof-of-principal for monitoring algal cultivation in a spatially-resolved manner, additional investment and effort is needed to advance the spatially-resolved technology from this initial demonstration to a robust field deployment test. For each proposed monitoring applications, a critical assessment of the tradeoffs between spatial and temporal resolution and field of view (FOV) needs to be performed to select the best spatially-resolved implementation (fixed platform/scanning/mobile platform, etc.) and optimize the scanning parameters for needs of different algal farm designs. For all the monitoring implementations discussed in this report, the sensing hardware is COTS; however, depending on the desired implementation there may be engineering and design required to integrate the sensor to the deployment platform (e.g. scissor-lift, UAV, etc.)

Outside of the optimal design and integration efforts, there are two significant challenges that need to be overcome to implement a spatially-resolved monitoring platform. First is developing a scheme for obtaining accurate estimates of the downwelling light at each spatial location. This will be especially important for scanning systems that cover large areas. Currently the success of the methods relies on the use of a downwelling fiber and spectrometer to measure the sunlight at a temporally matched time to the algal measurements. This works well because the ponds monitored are relatively small and near each other on the site. If ponds cover a larger area and/or are spread over a large area single downwelling spectral measurement is unlikely to be representative of all the areas due to variable cloud cover and shadows. It is likely cost prohibitive to install multiple downwelling spectrometers or to have a downwelling hyperspectral camera. However, we have thought of several options that could be tested.

- For mobile platforms (UAVs/drones) a single downwelling fiber/spectrometer could be mounted on the drone provided the payload was within the required SWaP. This would allow a near exact match of downwelling to upwelling spectra at every acquisition.
- For fixed/scanning units, it is more complicated. One option would be to simply monitor the intensity of light using an inexpensive light sensor at multiple locations through the area. The intensity measurements would be used to scale a known spectrum of sunlight to create downwelling spectra representative of smaller regions of interest throughout the area. This option assumes the spectrum of the sunlight is constant (which breaks down in heavily shadowed areas) and research and development would be needed to determine the optimal number of light sensors and temporal frequency of the data needed.

- An additional option would be to use a surface of opportunity (SOO) that is either naturally in the field of view of the sensor or placed there that has a known, broadband reflectance spectrum under solar illumination and could serve as a proxy for the sunlight. Again, research and development would be needed to determine suitable surfaces to fulfill this role and optimal placement and temporal frequency of the data needed.

The second challenge is around the data analysis. Acquiring data in a spatially-resolved fashion significantly increases the data output rate over the existing system. While it is possible that prediction of algal biomass could be made in near-real time, data fitting will likely involve an overhaul of the existing analysis strategies including a utilization of on board/edge computing architectures and possibly adopting artificial intelligence and machine learning approaches to augment the physics-based model to reduce computational workload. Our years of research has led to a well-characterized system with a myriad of field data (not laboratory) that could be used to train a model for prediction of algal biomass, and we have experience in this area for other hyperspectral sensing applications. However, this has not been done to date for monitoring algal cultivation.

It should be noted that a spatially-resolved spectroradiometric monitoring system has potential to provide key-information for attached algae systems (e.g. algal turf scrubbers) that can't be obtained through other methods. Attached algal systems present challenges with traditional methods for monitoring biomass because they are highly heterogeneous, dynamic, and typically contain more than one species. The spectral and spatial resolution afforded by spatially-resolved spectroradiometric monitoring could not only provide estimates of the biomass of the attached algal system, but also potentially distinguish the relative abundances of different species throughout the system. The latter would require some research and development to understand the sensitivity and specificity of the method.

Our team is very interested in working towards removing these barriers to large area, spatially-resolved scanning and welcomes future collaboration and projects in this area.

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